# CP-40 Series Close-Coupled Centrifugal Pumps

The CP-40 series is designed for clean water circulation in heating, cooling and pressure boosting systems. All wetted parts are made of stainless steel, which makes the pumps suitable for drinking water.

## Applications

- Heating and chilled water circulation in commercial buildings
- Pressure boosting in residential blocks
- Process water in light industry
- Irrigation from tanks and wells

## Technical data

### Hydraulic data

| Parameter | Value |
|---|---|
| Maximum flow | 24 m³/h |
| Maximum head | 46 m |
| Maximum operating pressure | 10 bar |
| Liquid temperature | −15 to +110 °C |
| Pipe connection | DN 40, PN 16 |
| Weight | 28 to 41 kg |

### Electrical data

| Parameter | Value |
|---|---|
| Motor power | 1.1 to 4.0 kW |
| Supply | 3 × 400 V, 50 Hz |
| Speed | 2900 rpm |
| Protection class | IP55 |
| Insulation class | F |
| Efficiency class | IE3 |

## Performance

| Model | Motor (kW) | Flow at best efficiency (m³/h) | Head at best efficiency (m) | Maximum head (m) | Current at 400 V (A) |
|---|---|---|---|---|---|
| CP-40-110 | 1.1 | 9.5 | 18.0 | 24 | 2.4 |
| CP-40-150 | 1.5 | 11.0 | 22.5 | 29 | 3.2 |
| CP-40-220 | 2.2 | 13.5 | 27.0 | 35 | 4.6 |
| CP-40-300 | 3.0 | 15.0 | 32.5 | 41 | 6.1 |
| CP-40-400 | 4.0 | 17.0 | 37.0 | 46 | 7.9 |

Performance values were measured with water at 20 °C in accordance with the tolerances of grade 2B. Curves for each model are available on request.

## Ordering code

| Field | Code | Meaning |
|---|---|---|
| Series | CP | Close-coupled centrifugal pump |
| Connection | 40 | DN 40 flanges |
| Motor | 110 | 1.1 kW |
| | 150 | 1.5 kW |
| | 220 | 2.2 kW |
| | 300 | 3.0 kW |
| | 400 | 4.0 kW |
| Shaft seal | B | Carbon and ceramic, standard |
| | V | Silicon carbide, for hot water |
| Material | S | Stainless steel 1.4301 |
| | T | Stainless steel 1.4404 |

For example, CP-40-220-V-S is a pump with a 2.2 kW motor, a silicon carbide seal and a casing in stainless steel 1.4301.

## Controller settings

Pumps with the optional VC-2 controller are configured with a plain text file. The default settings are:

```
[modbus]
address      = 12
baudrate     = 19200
parity       = even
stop_bits    = 1

[control]
mode         = constant_pressure
setpoint_bar = 4.5
ramp_up_s    = 5
```

Changes to the controller settings take effect after the controller has been restarted.

### Modbus registers

| Register | Name | Unit | Access |
|---|---|---|---|
| 40001 | Pressure setpoint | 0.1 bar | read/write |
| 40002 | Operating mode | — | read/write |
| 30001 | Actual pressure | 0.1 bar | read |
| 30002 | Motor current | 0.1 A | read |
| 30003 | Fault code | — | read |

## Installation notes

1) Install the pump with the motor shaft horizontal or vertical, never with the motor below the pump.
2) Fit isolating valves on both sides of the pump to allow servicing without draining the system.
3) Fill the pump and the suction pipe with water before the first start.

The pump must not run dry, not even for a short time.